

Technology Validation and Start Up Fund

Round 38 Proposal Evaluations

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TECHNOLOGY VALIDATION AND STARTUP FUND

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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 11 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the five-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (www.Redwdinnnov.com). Details of the TVSF program and the review process are provided in Appendix 2.

Twenty-three (23) TVSF Round 38 applications, totaling \$7,875,000, were received and evaluated.

The proposals consisted of 9 Phase 1 applications totaling \$5,175,000 (3 Phase 1, Option 1 applicants of \$975,000, 4 Phase 1, Option 2 of \$1,000,000 each, 2 Phase 1, Option 3 applicants of \$200,000) and 14 Phase 2 applications totaling \$2,700,000 for a combined total of \$7,875,000.

Funding is recommended for 8 Phase 1 applications totaling \$5,075,000 and 7 Phase 2 applications for a total of \$1,400,000 yielding a combined total of \$6,475,000. This translates to a 65% recommended application funding rate for Phase 1/Phase 2 TVSF Round 38, compared to the average of 51% over all 38 TVSF rounds. The recommended approval rate for Phase 2 proposals in this round is 50%.

2) Evaluation Results

Summaries of the evaluations of the proposals for Round 38, incorporating the interview, and funding recommendations are shown below in Tables 1 and 2. Round 38 interviews were conducted via video conference. The total recommended funding for Phase 1 and Phase 2 projects is, respectively, \$5,075,000, and \$1,400,000. Note that the Table 1 and 2 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Selection Committee which is weighted at 20 is four times as wide as project selection which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 1 and Phase 2 proposal may be found in Section 3 of this report.

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 1 – Phase 1 Proposal Evaluation and Funding Recommendation

TVSF Round 38

Proposal #	Applicant	Requested Funding (\$1,000)	Recommended Funding (\$1,000)	Selection Committee	Deal Flow & Budget Strategy	External Participation	Track Record	Metrics	Project Management & Experience	Project Selection Process
25-0363	Bon Secours Mercy Health Foundation	225	225							
25-0364	Case Western Reserve University	1,000	1,000							
25-0365	Cincinnati Children's Hospital Medical Center	250	250							
25-0366	Northeast Ohio Medical University	100	100							
25-0367	Parallax Advanced Research	100	0							
25-0368	The Cleveland Clinic Foundation	1,000	1,000							
25-0369	The Ohio State University	1,000	1,000							
25-0370	University Hospitals Health System	1,000	1,000							
25-0371	University of Cincinnati	500	500							
Total		5,175	5,075	Column width is proportional to score weighting in each category						
Evaluation Scale				Weak	Meets	Exceeds				

Option 1: \$200,000 to \$500,000, 1:1 cost share, 1 year to allocate funds, 2 year project period.

Option 2: \$1,000,000, 1:1 cost share, 2 years to allocate funds, 3 year project period.

Option 3: \$100,000, internship component, 1 year to allocate funds, 2 year project period.

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 2 – Phase 2 Proposal Evaluation and Funding Recommendation

TVSF Round 38

Proposal Number	Lead Applicant	Requested Funding (\$1,000)	Recommended Funding (\$1,000)	Team	Opportunity/Market Size	IP Protection	Proof of Concept	Potential Investor & Business Partner Engagement	Business Model	Project Plan/Budget Narrative	Growth Plan in Ohio	ESP Interaction
FY25-0373	Astute Signals, Inc	200	200									
FY25-0374	Bowerbags LLC	200	200									
FY25-0375	BrandRank.ai	200	200									
FY25-0376	Dayton Photonics Systems	200	200									
FY25-0377	DEB Technologies, LLC	200	200									
FY25-0378	Galvanix	200	200									
FY25-0379	Ignite Assurance Platform	200	200									
Sub-Total		1,400	1,400									
FY25-0372	A.M. ToolBox LLC	200	0									
FY25-0380	Impeccable Air	100	0									
FY25-0381	Innovision, LLC	200	0									
FY25-0382	Parcell Company	200	0									
FY25-0383	SimWerx, Inc	200	0									
FY25-0384	Skuld LLC	200	0									
FY25-0385	Unspent	200	0									
Sub-Total		1,300	Column width is proportional to score weighting in each category									
Total		2,700										
Evaluation Scale			Weak	Meets	Exceeds							

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 3 lists the funding approval rate by TVSF round. This round's approval rate of 65% of the total submitted proposals is on the high side of the average over all rounds. The Phase 1 applications recommended for approval comprised 35% of the submittals for this round. The historical range of recommendations for individual rounds has spanned 27 – 100%, with an average of 51%.

Table 3. TVSF Approval Rate by Round

TVSF Round 38 Approval Rate by Round Phase 1/Phase 2

Round	\$ Recommended	Approval Rate
1 (APR 2012)	\$950,000	35%
2 (AUG 2012)	\$900,000	52%
3 (DEC 2012)	\$610,000	44%
4 (JUN 2013)	\$864,000	30%
5 (FEB 2014)	\$1,462,000	46%
6 (JUN 2014)	\$998,000	39%
7 (OCT 2014)	\$1,100,000	57%
8 (FEB 2015)	\$710,000	37%
9 (JUN 2015)	\$550,000	31%
10 (DEC 2015)	\$925,000	38%
11 (APR 2016)	\$1,239,000	46%
12 (OCT 2016)	\$3,537,269	46%
13 (MAR2017)	\$1,567,500	38%
14 (SEP 2017)	\$498,832	27%
15 (DEC 2017)	\$2,250,000	38%
16 (MAR 2018)	\$2,098,600	52%
17 (SEP 2018)	\$2,100,000	42%
18 (DEC 2018)	\$1,150,000	35%
19 (APR 2019)	\$2,250,000	43%

Round	\$ Recommended	Approval Rate
20 (NOV 2019)	\$1,350,000	43%
21 (FEB 2020)	\$3,944,000	56%
22 (JUN 2020)	\$1,398,630	53%
23 (DEC 2020)	\$900,000	50%
24 (MAR 2021)	\$2,092,900	55%
25 (JUN 2021)	\$800,000	75%
26 (OCT 2021)	\$1,700,000	55%
27 (FEB 2022)	\$850,000	43%
28 (APR 2022)	\$2,499,976	64%
29 (JULY 2022)	\$850,000	100%
30 (OCT 2022)	\$3,700,000	71%
31 (JAN 2023)	\$100,000	50%
32 (APR 2023)	\$850,000	64%
33 (JULY 2023)	\$1,100,000	73%
34 (OCT 2023)	\$250,000	33%
35 (JAN 2024)	\$800,000	59%
36 (APR 2024)	\$2,350,000	80%
37 (JULY 2024)	\$1,798,000	82%
38 (OCT 2024)	\$6,475,000	65%
Total Funding	\$ 59,568,706.94	
Average/Round	\$ 1,567,597.55	51%

3) Proposal Summaries

Phase 1 Proposal Summaries - Recommended for Funding

Proposal 25-0363	Bon Secours Mercy Health Foundation	Amount Requested: \$225,000
Prior Phase 1 Applications: Yes	Technology Validation Fund #4 -Bon Secours Mercy Health Foundation	Amount Recommended: \$225,000

TOTAL BUDGET: \$450,000

Institution Snapshot: Technology Validation Fund #4 (BSMH-TVF#4) is a pool of funds designed to formalize a commercialization process to advance new products and ideas originating from within the Bon Secours Mercy Health (BSMH) system.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Selection Committee	The Selection Committee members listed have served in the same capacity on other Funded rounds. Good balance of entrepreneurial, startup, technology validation and commercialization process backgrounds.
Y	Deal Flow; Budget Strategy	It is anticipated that the Selection Committee will receive 5 to 7 technologies to consider for funding of 3 to 4 projects. The estimated validation cost is \$100K to \$150K per project.
G	External Participation	Application clearly recognizes the importance of 3rd party providers and provides a credible list of prospective 3rd party firms. Third party external providers are required for each funded project.
Y	Track Record	Five of eight projects from 3 previous awards have been completed and are in licensing discussions or identifying next steps in commercialization.
Y	Metrics	Metrics have yet to be tested to maximum extent. Currently slated to have 1-2 Newcos and 1-2 licenses from the fund. No Newcos formed or licensed signed to date from previous funds.
G	Project Management & Experience	The current project team has not changed since last round of funding. The management of the projects is standardized, license transactions are handled by an outsourced company.
G	Project Selection	Project selection process is clear and well described.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: Bon Secours has received 3 previous Phase 1 awards which have funded a total of eight projects. The Innovation team has performed the initial diligence for the projects under consideration and the committee has selected the projects to fund. The commercialization has been handled by an outsourced company for the 3 previous awards. This is being critically reviewed to determine if that function will be brought into the Innovation team.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0364	Case Western Reserve University	Amount Requested: \$1,000,000
Prior Phase 1 Applications: Yes	Phase 1 -Technology Validation Start-Up Fund - Pool of Funds	Amount Recommended: \$1,000,000

TOTAL BUDGET: \$2,000,000

Institution Snapshot: The Phase 1 -Technology Validation Start-Up Fund-Pool of Funds will be used to advance technology generated within the organization to a point where a startup can obtain external 3rd party funding.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	There are 14 members from relevant educational and experiential backgrounds. Each member has significant startup, industry and venture capital expertise to help recruit strategic partners and entrepreneurs to aid commercialization of funded projects.
G	Deal Flow; Budget Strategy	Deal flow is over 150 invention disclosures per year. CTP funds technologies pre-vetted in other funding programs; selects those likely to be commercialized. Budget is consistent with 2-year grant.
Y	External Participation	CWRU has a long list of 3rd party vendors that provide services for them on CTP and other programs as well.
G	Track Record	CWRU has an impressive track record of both NewCo creation and licensing. In addition, there is significant follow-on funding for 15 startups of over \$197M and 3 of the 15 are publicly traded.
G	Metrics	Overall CWRU commercialization metrics and aspirations are strong. Specific metrics for this TVSF application were not provided.
G	Project Management & Experience	The experience of the people and management of the projects is well established and effective.
G	Project Selection	There is a clear, detailed selection process for this Fund.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: CWRU continues to improve the TVSF Phase 1 Fund program. The changes to this proposal include implementing a strategic planning program, C3i, to help applicants with educational sessions, mentorship teams and commercialization planning services to help them develop an effective application. The second change is to split the healthcare and non-healthcare into two different application cycles. Overall, CWRU is running a very effective Phase 1 TVSF program.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0365	Cincinnati Children's Hospital Medical Center	Amount Requested: \$250,000
Prior Phase 1 Applications: Yes	<i>Cincinnati Children's Innovation Ventures Acceleration Program</i>	Amount Recommended: \$250,000

TOTAL BUDGET: \$500,000

Institution Snapshot: Cincinnati Children's Innovation Ventures Acceleration Program will be used for the needed resources, expertise, and infrastructure to progress technologies with commercial potential until they are commercialization ready.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	A well rounded 5-member committee with skills and experience in life sciences, start-ups, commercialization of innovations, fund raising and business development.
G	Deal Flow; Budget Strategy	CCIV has identified four distinct focus areas and has sustained appropriate deal flow for a long time.
G	External Participation	An expansive network of external providers is used for project validation activities. OH-based 3rd party vendors are stressed and used by funded projects.
Y	Track Record	Cincinnati Children's has a track record of commercializing intellectual property by creating new or partnering with existing companies.
G	Metrics	Reasonable categories of leading indicator metrics are provided, specific goals are not. Outcome metrics are 2-3 new start-ups/licenses emerging from the funding of 3-4 projects.
G	Project Management & Experience	Awardee is assigned a dedicated CCIV team member who serves as a commercialization coach, providing dedicated attention to the success of the technology.
G	Project Selection	Selection process in use is reasonably standard for these awards.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: Cincinnati Children's has a well-established process for selecting appropriate candidate technologies for the TVSF Phase 1 program. There is sufficient deal flow to carefully choose candidates that might initiate a start-up company in Ohio. A CCIV team member is assigned to each team to serve as a commercialization coach and OH-based vendors are emphasized to the project teams. Metrics are consistent with the careful selection of projects.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0366	Northeast Ohio Medical University	Amount Requested: \$100,000
Prior Phase 1 Applications: Yes	<i>Validation of Biomedical Technologies at Northeast Ohio Medical University</i>	Amount Recommended: \$100,000

TOTAL BUDGET: \$100,000

Institution Snapshot: The pool of funds is to support medical innovation and technologies developed by NEOMED faculty, staff, or students in the areas of device, drug, or drug delivery development.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Selection Committee	The 5-member team has complementary backgrounds in areas needed to select technology for commercialization. The Chair is a serial entrepreneur with 5 startup experiences.
G	Deal Flow; Budget Strategy	Deal flow approach is well considered and reasonable for the requested TVSF award.
Y	External Participation	The use of external 3rd party providers is encouraged, not mandated. 'Selection/Oversight committee will . . . provide information to project leads.'
Y	Track Record	2 projects funded in the 2 Phase 1 programs to date. 1 has 1 patent approved, 1 issued, SBIR funding, negotiating license to inventor's company. Second project not completed prior to inventor departure.
R	Metrics	Metrics tracked are being developed by funded team and selection committee. The organization is relatively new to technology transfer.
Y	Project Management & Experience	Selection committee has strong entrepreneur and project management backgrounds. Projects are managed by individual PIs. No specific oversight is noted for project guidance.
Y	Project Selection	A detailed selection process is outlined in the application.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: NEOMed has a selection committee with complementary backgrounds. The Chair is a serial entrepreneur of 5 startups. Program manager is highly motivated and well connected with the innovation community. Only 2 projects have been funded from previous Phase 1 funds. One successful and one not completed. The program is relatively young and metrics are not well established. Benchmarking with similar sized organizations may help the team better leverage the success of others in technology commercialization.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0368	The Cleveland Clinic Foundation	Amount Requested: \$1,000,000
Prior Phase 1 Applications: Yes	<i>Cleveland Clinic Technology Validation and Start-up Fund VI (TVSF-CC 6)</i>	Amount Recommended: \$1,000,000

TOTAL BUDGET: \$2,000,000

Institution Snapshot: The TVSF-CC Program fast-tracks technology development, validation and the ultimate spinning out of technology developed at The Cleveland Clinic Foundation.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	The 4 voting members of the selection committee have extensive complementary experience in startups, venture capital, entrepreneurship, technical and fundraising. It is well aligned with focus areas.
G	Deal Flow; Budget Strategy	The Clinic receives more than 200 new opportunities per year. Budget strategy is to get the program to the next level for licensing or startup.
Y	External Participation	Process in place to determine use of internal or 3rd party resources, which is ultimately decided by the Selection Committee.
Y	Track Record	19 projects started since 17-066 with 7 resulting licenses and 1 acquisition. 2 companies formed. 13 projects currently underway from award 22-1692 with no licenses yet
G	Metrics	Projects awarded, licenses executed, Ohio startups, follow-on funding are some of the metrics tracked. Estimate 8 projects will receive funding with 7 resulting licenses.
G	Project Management & Experience	The Innovations Development director assigned to the TVSF project will oversee that specific individual project on a day-to-day basis to ensure the project is on track.
G	Project Selection Process	A comprehensive overall process was described.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: The Cleveland Clinic has significant deal flow to consider for the TVSF Phase 1 program. The Selection committee is strong with complementary skill sets and experiences. There is project management oversight to keep the project on track and outside vendors are encouraged to be used wherever possible.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0369	The Ohio State University	Amount Requested: \$1,000,000
Prior Phase 1 Applications: Yes	<i>Accelerator Awards</i>	Amount Recommended: \$1,000,000
TOTAL BUDGET: \$2,000,000		

Institution Snapshot: The strategic vision of the Accelerator Awards program is to fund translation of cutting-edge Ohio State innovations and accelerate them into the marketplace through the creation of Ohio-based startup companies.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	The 18-member selection committee is comprised of entrepreneurs, technical experts, investors/venture capital and local ESP representatives representing multiple disciplines and areas of expertise.
G	Deal Flow; Budget Strategy	There are an average of 370 invention disclosures per year across 15 colleges. The current requested budget was demonstrated as adequate in the last round of funding received by OSU.
Y	External Participation	Seventy five percent of the allotted funds must be used for external vendors. Up to 25% may be used for activities that support in-lab work and personnel.
G	Track Record	96 projects funded, 16 in progress. Of 80 remaining, 24 were non-viable (30%), 56 are still in commercialization pipeline with 25 licenses, 1 option. \$110M in follow on funding raised by 23 start-ups.
Y	Metrics	Success and impact metrics are tracked and include number of technologies funded, projects completed and outcomes, licenses executed, follow-on funding, etc.
Y	Project Management & Experience	A cohort- based model with 2 application cycles per year has clear quarterly reporting requirements to maintain funding and monitor progress of project.
G	Project Selection	Clear process for project selection was detailed in the application.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: OSU has a large selection committee comprised of entrepreneurs, technical experts, local ESP representatives and investors. The deal flow of ~370 invention disclosures per year represents a rich source of potential projects. To date 96 projects have been funded resulting in 25 licenses with \$110M in follow-on funding raised by 23 start-ups. Applicants are required to engage an industry expert prior to full proposal.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0370	University Hospitals Health System, Inc.	Amount Requested: \$1,000,000
Prior Phase 1 Applications: Yes	<i>University Hospitals Technology Venture Fund</i>	Amount Recommended: \$1,000,000

TOTAL BUDGET: \$2,000,000

Institution Snapshot: University Hospitals Technology Venture Fund earmarks a pool of funds with the purpose of de-risking and accelerating the commercialization of intellectual property (“IP”) created or reduced to practice within UH.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	The 12-member selection committee has experience and expertise in investing, health care entrepreneurship, clinical innovation and financing.
G	Deal Flow; Budget Strategy	It is anticipated that 40-50 eligible technologies will be considered annually by UHTVF and 9-12 will be funded over the 2-year project timeframe.
G	External Participation	ESPs, local 3rd party companies are used to validate technology and development plan. Have a vendor database that lists past performance. Also provide an extensive list of prior providers.
Y	Track Record	From the first 2 funds, 4 of 14 projects have been licensed to startups. 3 others are in negotiations from Fund 3 and 4.
Y	Metrics	UH tracks a long list of metrics from EIR hours to licenses, jobs created and regulatory milestones. Estimate 3-6 licenses from 9-12 Fund V projects licensed to new or early-stage Ohio companies.
G	Project Management & Experience	UHV has developed management systems to keep track of project related and technology specific details. Admin management team meets monthly with project teams.
G	Project Selection Process	A detailed 4 stage gate program is in place.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: University Hospitals has a well established TVSF Phase 1 program with a well-balanced selection committee, project selection process and program management. The expected outcome is that 9 to 12 projects will be funded over the 2-year period and 3-6 licenses will be signed with Newcos or early-stage Ohio companies.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0371	University of Cincinnati	Amount Requested: \$500,000
Prior Phase 1 Applications: Yes	<i>University of Cincinnati TVSF Phase 1 20248</i>	Amount Recommended: \$500,000

TOTAL BUDGET: \$1,000,000

Institution Snapshot: The University of Cincinnati uses the TVSF Phase 1 funds to accelerate qualified UC intellectual property into new ventures.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	Selection committees are custom configured for each opportunity. Consistency is provided by an interview process and the ongoing presence of Director and / or Assistant Director of Ventures lab.
Y	Deal Flow; Budget Strategy	The current pipeline suggests that 12-15 accelerator projects will pitch for funding and 6 to 8 will receive Phase 1 funding. The last funded proposal took more than a year to commit funds.
Y	External Participation	There is a long list of potential external providers for this program. ESPs and other Accelerator programs are used before TVSF Phase 1 dollars are used.
Y	Track Record	2019 to 2022 there were 23 teams funded and 7 licenses issued since 2019 and approximately 69 jobs created. Frequent staff turnover has made it difficult to capture the historical perspective.
Y	Metrics	Metrics tracked include teams funded, IP licenses, options and jobs created.
Y	Project Management & Experience	Projects are led by mutually agreed upon EIRs from a pool of > 80. EIRs are responsible day to day and meet with Accelerator Leadership quarterly.
Y	Project Selection	Selection process is clear and well considered.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: The selection committee is tailored to experts within that technology field. This provides early connections to funding organizations and potential corporate partnerships. The pipeline is adequate for the deal flow anticipated. Frequent staff turnover has made it difficult to capture the historical track record. Projects are led by EIRs generally from industry and they are responsible for the day to day running of the organization.

Proposal Summaries - Phase 1 Not Recommended for Funding

Proposal 25-0367	Parallax Advanced Research	Amount Requested: \$100,000
Prior Phase 1 Applications: No	Parallax TVSF Research Commercialization Fund	Amount Recommended: \$0

TOTAL BUDGET: \$0

Institution Snapshot: The Parallax TVSF Research Commercialization Fund will be used to select projects that can leverage non-dilutive dollars to establish critical early-stage milestones that are established based on customer and investor input.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	The 4-person selection committee has significant experience in commercializing technologies from research institutions. In addition, fund raising, successful start-up creation and investment in startups are skills integral to the members.
Y	Deal Flow; Budget Strategy	Deal flow and project budget strategy are adequate for the amount of funding requested.
Y	External Participation	A strong statement is provided about the use of third-party vendors. No prospective vendors are mentioned.
Y	Track Record	Parallax took over Wright State Applied Research Corporation Phase 1 award midway through. Manager of this award managed that Phase 1 for 15 months; 1 of 2 projects funded received Phase 2
Y	Metrics	Parallax will track # of projects funded, resulting commercialization activity and other metrics requested by ODOD.
R	Project Management & Experience	There is no TTO function within Parallax at this time. Projects will be assigned an EIR from the ESP. In addition, the program manager will also manage the individual projects.
Y	Project Selection Process	Selection process is clear and appropriate for requested funds size.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: The TVSF program requires the applicant to have an operating Technology Transfer Office (TTO). Parallax does not have a TTO function within the organization at this time and therefore, is not an eligible applicant for this round. The selection committee has experience and skills appropriate for choosing projects to be funded. Parallax is working closely with the regional ESP and the ESP will provide an EIR to coach the project teams. Parallax was selected to run the previous program when the Wright State Applied Research Corporation contract was not renewed.

Phase 2 Proposal Summaries - Recommended for Funding

Proposal 25-0373	Astute Signals, Inc	Amount Requested: \$200,000
<i>Licensing Institution</i>	University Hospital Health System Inc	Amount Recommended: \$200,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	<i>Astute Signals</i>
Biomedical/ Life Sciences		JumpStart / Entrepreneurs' Center

Company Snapshot: Astute uses AI to analyze facial expressions, vocal tone, and body language to identify potential signs of mental health conditions and assist in initial screening of depression and anxiety disorders.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Highly qualified technical core team of experienced serial entrepreneur physicians with successful track record in AI company growth and telemedicine. Highly experienced Board.
Y	Opportunity/Market Size	Assuming a fee of \$2 per screening, initial total addressable market is \$872M.
Y	Intellectual Property Protection	Proprietary algorithm and database, no patents. In negotiation with University Hospitals for license.
G	Proof of Concept	Successful proof of concept study in Emergency Department demonstrated app could collect video recordings for signs of depression and anxiety. IRB approval is pending for clinical study.
G	Potential Investor/ Business Partner Engagement	Advanced conversations with investors. UH Ventures committed funding to this effort and will provide \$200K of non-dilutive funding.
G	Business Model	Telehealth first market. Licensing model well described. Product does not require clinical trial or prior FDA approval for sale.
Y	Project Plan/ Budget Narrative	Tasks & milestones clearly presented & achievable in funding period.
Y	Growth Plan in Ohio	Company based in Cleveland. Anticipate 15-20 FTEs by 2028.
G	ESP Interaction	JumpStart and Dayton Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Experienced team including serial entrepreneur physician founders. Large market opportunity accessible without requiring prior FDA approval for sale. First of its kind system collecting multiple data streams (visual, vocal, etc.) has been successfully demonstrated. Possible platform for many conditions. Strong investor relationships with commitments for funding. This application is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0374	Bowerbags LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Army and Airforce Labs	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 30	Load Distribution System (LDS)
Biomedical/ Life Sciences		Entrepreneurs' Center

Company Snapshot: Bowerbags is integrating Army-developed load distribution systems with AFRL technology to create modular tactical solutions to enhance performance and mobility while reducing injury rates through better ergonomic weight load management.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Team has the needed experience and connections in the target markets. The team is the same as that of the previous application of Round 30.
Y	Opportunity/Market Size	The TAM is over \$1 Billion per year with 16% CAGR.
Y	Intellectual Property Protection	Negotiating licensing of US 9,700,122B2. This should complement the earlier licensed patent, US 8,857,681, in earlier application to develop a viable product.
Y	Proof of Concept	TRL is 8 for the LDS technology based on Army testing.
Y	Potential Investor/ Business Partner Engagement	Has been in contact with potential business partners. Too early for investor or partnership agreements.
Y	Business Model	A 3-prong approach with defense, first responder and recreational markets.
Y	Project Plan/ Budget Narrative	Working with contract service providers with one part time staff.
Y	Growth Plan in Ohio	Project 30 FT staff in 2027.
Y	ESP Interaction	The company is interacting with the regional ESP.
	Evaluator Recommendation	This application is recommended for funding

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The Team is focused on improving the performance of the integrated unit based on the lessons learned from the unsuccessful attempt in Round 30. Current effort is targeted at integrating the LDS technology with their device for use in wider range of ergonomics patents to meet the updated specs of the Army. The Team is urged to focus on demonstrating the applicability and acceptance of the improved device and work toward commercializing the products in the DoD and first responder markets.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0375	BrandRank.ai	Amount Requested: \$200,000
<i>Licensing Institution</i>	National Security Agency	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Anti-Hallucination Filter for Generative AI Answer Engines</i>
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: BrandRank.ai aims to create a critical new offering: an anti-hallucination filter to verify accuracy in user exchanges with any answer engine.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	The founders and the rest of the team's backgrounds are appropriate and impressive.
G	Opportunity/Market Size	Ensuring the integrity and accuracy of a website's interactive content is a large and rapidly growing market.
Y	Intellectual Property Protection	The team is currently negotiating for rights to an NSA patent that expires in 2034. A provisional patent has been filed to extend and strengthen coverage.
G	Proof of Concept	The base software is TRL 6 and the anti-hallucination capability is at TRL 3-4. The proposed TVSF proof of concept is well considered.
G	Potential Investor/ Business Partner Engagement	The team has strong engagement with investors and initial adopters including the Better Business Bureau and Nestle.
G	Business Model	A SaaS product platform to measure brand attributes is the basis for the business. Anti-hallucination capability will be a new feature.
G	Project Plan/ Budget Narrative	The project plan is well constructed to move TRL from 3-4 to 6.
G	Growth Plan in Ohio	The company intends to grow in Ohio, creating 30 jobs by 2029.
Y	ESP Interaction	Significant interaction with the Entrepreneurs Center is mentioned.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The BrandRank.ai application is recommended for funding. It targets the large and growing market of ensuring the integrity of information provided by interactive websites. The team is well-qualified and the proposed project is well considered.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0376	Dayton Photonics Systems LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Army Research Laboratory	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Fiberless Laser Communication
Sensors	Entrepreneurs' Center	

Company Snapshot: Dayton Photonics Systems LLC 's mission is to provide fiberless optical communication as a new tool to solve The United States broadband infrastructure challenges in 4-to-10-mile range.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	CEO and CTO have suitable experience, background and track records in photonics development. Excellent working relationships with AFRL and local fund-raising organizations.
G	Opportunity/Market Size	TAM market size is in the billion \$/yr range with >30% CAGR.
Y	Intellectual Property Protection	Negotiating for exclusive license in the field of use for U.S. Patent 9,932,254 issued in 2017.
Y	Proof of Concept	TRL 6 for < 4 miles range with SBIR funding. TRL 4 for the target 10-mile range and 10 GBS.
G	Potential Investor/ Business Partner Engagement	Early- stage discussions with an investment of \$1M secured in Aug 2024.
Y	Business Model	Credible model with sales of 2-component links & SaaS as revenue sources. Projected revenue of \$0.16M & \$84M in 2024 & 2028.
Y	Project Plan/ Budget Narrative	Timeline is reasonable to obtain MVP in 12 months.
Y	Growth Plan in Ohio	Projects the creation of > 50 jobs.
Y	ESP Interaction	Working with local ESP and was guided to potential investors.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: A very strong technical team with active fund raising and business development skills. The CEO has secured one million \$ investment recently from a venture group. The market opportunity for the mid and small tier ISP markets is large and the offerings from Dayton Photonics seem to have competitive advantages to capture significant share of the emerging market. The TEAM is encouraged to demonstrate commercial feasibility in the small tier market quickly and enter the larger mid-tier market with needed fund raising and partnership engagements. The application is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0377	DEB Technologies, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Department of Veterans Affairs	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 37	DEB Technologies, LLC
Biomedical/ Life Sciences		Entrepreneurs' Center

Company Snapshot: This magnet-based technology is a Device for gait, Efficiency, and Balance (DEB) Shoe to prevent leg crossing and subsequent falls in patients with limited leg strength.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	CEO experienced in procurement, company and product development. Track record of successful exit of a medical supply chain management company. Inventor is part-time consultant.
Y	Opportunity/Market Size	Market opportunity is approximately \$95M.
Y	Intellectual Property Protection	Patent application licensed from VA 26 July 2024.
G	Proof of Concept	Pilot study at the VA demonstrated ability to improve from no walking to walking up to 400 feet with improved overall safety.
G	Potential Investor/ Business Partner Engagement	Discussions with venture capital & Ohio Angel investors. Committed lead investor for seed round. Have begun discussions with others.
Y	Business Model	Company has access to VA Centers through VA Intrapreneurship Program. Class 1 Medical Device with clear path to sales.
G	Project Plan/ Budget Narrative	Schedule and budget are well developed. Sole source provider used.
Y	Growth Plan in Ohio	Company promises to remain headquartered in Ohio.
G	ESP Interaction	Working closely with Dayton Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Leadership team has experience in technical and medical business development. Strong team of strategic partners and advisors. Moderate market addressable with low manufacturing costs, preferred access to customers and low regulatory burden. Early access to customers via the VA's Intrapreneurial Product Market Place. Technology demonstrated in VA rehab setting. Planned work at established Ohio product development firm will result in manufacturable prototypes. This application is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0378	Galvanix Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Case Western Reserve University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Carbon-free, Domestic Rare Earth Metal Production</i>
Advanced Manufacturing		Rev1/ Entrepreneurs' Center/TechGROWTH

Company Snapshot: Galvanix is commercializing a novel process for producing rare earth metals that only generates the desired metal, oxygen, and low-grade heat.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The team has direct technical experience and relevant business experience. Fund raising experience is currently lacking, in the short term they will use support provided by the ESP and are targeting a new hire in the first 6 months.
Y	Opportunity/Market Size	Initial focus on neodymium extraction; technology is innovative, provides environmental benefits and can be extended to other materials. US market size supported by customer feedback.
Y	Intellectual Property Protection	Options for 2 filed US patents and 1 international patent in discussion, 2 additional patents in progress. Future patents may be developed, competitor patents will require further evaluation.
Y	Proof of Concept	Lab scale proof PoC will be extended to commercial demonstration scale system. Material will demonstrate MVP based on critical metrics identified by potential customers and investors.
Y	Potential Investor/ Business Partner Engagement	Current support has come from founder capital. Federal funds for future development, preliminary discussions with investors initiated.
Y	Business Model	Initial operation will offer toll manufacturing with future extension to a technology development and licensing model.
Y	Project Plan/ Budget Narrative	Budget is consistent with demonstrating critical PoC within 12 months
Y	Growth Plan in Ohio	"Deep tech" start-up using CWRU expertise, 20 jobs in next 5 years.
Y	ESP Interaction	The Entrepreneur's Center, Dayton and TechGROWTH.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Galvanix has demonstrated innovative, patented, platform technology for the extraction of critical materials used in rare earth magnets that are pivotal to the growth of environmentally friendly energy resources. Feedback from potential domestic commercial customers and investors has focused the short-term business actions; continued close interaction with these supporters will be extremely valuable to achieve the business potential. Additional expertise in fund raising is available for short term involvement and will need to be brought into the business to meet the projected growth plans.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0379	Ignyte Assurance Platform	Amount Requested: \$200,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$200,000
Prior Phase 1 Applications:	Prior Phase 2 Applications:	AI Enabled Cyber Risk Platform
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: Ignyte Assurance Platform aims to integrate advanced AI technology into a cybersecurity platform to protect organizations from growing cyber threats.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The team has matured and grown since their last, unrelated application in 2022. A former team member is now an advisor / channel partner at CDW, and a new team member is focused on channel partner acquisition.
G	Opportunity/Market Size	Protecting secure data from AI enabled threats is a large and growing market. Ignyte is focusing on defense contractors who wish to comply with the most demanding certification of FedRAMP.
Y	Intellectual Property Protection	Ignyte is in conversations to license a new USAF patent application "Cross Domain Solution Architecture".
Y	Proof of Concept	The new technology is at TRL 5/6. The TVSF funding will allow advancement to TRL 7 (Demonstration in Operational Environment).
Y	Potential Investor/ Business Partner Engagement	Substantial engagement is described with early adopters and prospective investors.
Y	Business Model	The team has adopted a SaaS business model largely implemented through channel partners like CDW.
Y	Project Plan/ Budget Narrative	The project plan is clear and reasonable.
Y	Growth Plan in Ohio	The company is clearly committed to grow in Ohio.
G	ESP Interaction	The application describes a high level of ESP engagement.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: This application is recommended for funding. The team has continued to learn and implement product improvements that address the AI driven challenges / opportunities in the Federal data security market. This application is distinct (both product offering and IP licensed) from prior Round 27 and 28 applications.

Proposal Summaries - Phase 2 Not Recommended for Funding

FY25-0372	A.M. ToolBox LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Naval Surface Warfare Center - Carderock	Amount Recommended: \$ 0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Low Density Ceramic Armor Material</i>
Advanced Materials		Entrepreneurs' Center

Company Snapshot: A.M. ToolBox LLC is proposing to develop and market a lower cost, lighter weight ballistic protection body armor while maintaining high user agility.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	One member has experience in AM technology, two members are listed as attorneys. Limited experience in design of armor components provided. No experience in transition from service base to business supplying products.
R	Opportunity/Market Size	US market size for body armor identified. TAM for the first responders and private security is not given. Development combines novel material with AM to produce complex shaped protective armor.
R	Intellectual Property Protection	Negotiating exclusive rights to Patent #8105510 expiring ~ November 2030. License covers use for private security and first responders only. Manufacturing technology expertise based on experience.
R	Proof of Concept	Base material has demonstrated TRL 6, MVP for Bio-metric curve product should be available within 12 months. No evidence of customer input on metrics, stated metrics are general market forces.
R	Potential Investor/ Business Partner Engagement	No details of discussions with investors provided.
R	Business Model	Traditional sales & marketing strategy. No plan to address barriers to entry. Development relies on Phase 2 SBIR, Phase 1 not yet started.
Y	Project Plan/ Budget Narrative	Plan/budget are aggressive. ~50% of funds for new equipment.
Y	Growth Plan in Ohio	Indicate 25 jobs, not clear which company and where it will be.
Y	ESP Interaction	Stated interaction with The Entrepreneurs' Center.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The proposed work combines two novel technologies to provide personal protective armor that can match the contours of the human body. If successful, the product line will enter a market that has significant competition from existing companies. Long term success for a new venture in the target market requires the proposers transition from a service-based business to one providing products, obtain external funding to expand routine manufacturing and develop a sales and marketing approach to compete effectively with established companies in the selected market segment.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0380	Impeccable Air	Amount Requested: \$100,000
<i>Licensing Institution</i>		Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Impeccable Air Start-Up & Expansion</i>
Advanced Materials		

Company Snapshot: Impeccable Air has developed a two-piece vent system with a unique filter mechanism to provide an attractive and effective whole-home air purification solution.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	One person Team—the founder, CEO. Need to add other skill sets/ capabilities to realize the potential of the technology / products.
R	Opportunity/Market Size	Market size for HEPA filters is in the billion-dollar range and growing. Size of the market for residential air filters is not mentioned. Market is dominated by several world class large competitors.
R	Intellectual Property Protection	3 patent applications and one Trade Secret—non from organizations recognized by the State as eligible institutions.
Y	Proof of Concept	Early field trials at 60 homes shows superior performance. Has been getting positive responses from potential customers/ distributors at trade shows.
Y	Potential Investor/ Business Partner Engagement	Has initiated discussions with Home Depot, Orbis Partners and others. Interest seems to be quite high.
Y	Business Model	Focused on initial sale of filtration unit and subscription model to sell high efficiency patented filter for \$4/unit every 3 months.
Y	Project Plan/ Budget Narrative	Well laid out plan. Seeking \$100K funding \$190K project.
Y	Growth Plan in Ohio	Yes, plans to build and grow the business from OH.
R	ESP Interaction	Youngstown State University mentioned; not an ESP.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The application does not meet the State requirements for IP source as well as ESP engagement. Funding request is not recommended.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0381	Innovision, LLC.	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of California, Santa Cruz	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Ufit-Exercise
Biomedical/ Life Sciences		Entrepreneurs' Center

Company Snapshot: Innovision, LLC. will develop a hardware-software integrated system, **Ufit-Exercise**, to create a synthetic, immersive, and interactive environment for physiotherapists and patients to conduct Physiotherapy Scoliosis Specific Exercises (PSSEs) effectively.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Technically strong team with some external support for regulatory and software. Insufficient Business development, sales and fundraising expertise.
Y	Opportunity/Market Size	Market opportunity exceeds \$400M.
Y	Intellectual Property Protection	One patent application from Innovision and one from University of California Santa Cruz. Letter of support indicates willingness to license exclusively if TVSF is funded.
G	Proof of Concept	Basic system capabilities demonstrated, including Microsoft Kinect motion capture, patient digital twin creation, and 3D replication of physical therapy exercises and tracking.
R	Potential Investor/ Business Partner Engagement	Virtually no contact with potential investors.
R	Business Model	Sales goals unrealistic. No sales expertise. Regulatory strategy needs validation. Competitive strategy is unclear.
R	Project Plan/ Budget Narrative	Plan & milestones clearly presented. Unclear how co will fund project.
Y	Growth Plan in Ohio	Principals live in Ohio and have committed to company growth here.
Y	ESP Interaction	Regular meetings with Dayton Entrepreneurs' Center.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Leadership team is strong technically but lacks business development, sales and fundraising. Intellectual property meets basic requirements but will need to be strengthened with future IP and/or trade secrets. Regulatory strategy needs validation. Business model lacks strategy to sell and to compete with similar FDA-approved products already on the market. Funding needs are not met with TVSF alone, and gap funding strategy is not well described. This application is not recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0382	Parcell Company	Amount Requested: \$200,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Integrated Tracking System</i>
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: Parcell Company provides a secure data communication capability for industrial hardware devices, enhancing both efficiency and cybersecurity.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Experienced team that has raised non-dilutive capital since Round 24 award for a different product / market. Plan to add DoD sales expertise in 2025.
R	Opportunity/Market Size	Team is focused on streamlining USAF secure tool management process. Market is small at \$24M, hesitant to pay for value.
Y	Intellectual Property Protection	In negotiations with USAF for US 10,901,928 to enhance data security and compliance.
Y	Proof of Concept	Initial demonstrations have occurred at MacDill and Nellis Air Force Bases.
Y	Potential Investor/ Business Partner Engagement	Parcell has made significant penetration into USAF user and decision-making communities. Adept at securing SBIR funding.
R	Business Model	No discussion of pro-forma was provided in application. Interview discussions were not encouraging around USAF paying for value.
Y	Project Plan/ Budget Narrative	Project plan is reasonable and creditable.
G	Growth Plan in Ohio	A clear commitment to growing in Ohio is expressed.
G	ESP Interaction	Significant engagement with the Entrepreneurs Center is described.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Parcell received funding in Round 24 for a secure home parcel delivery container. In this round, Parcell is focused on an important USAF need – secure and efficient management of aircraft maintenance tools. Unfortunately, the market as described is quite small and does not appear to be willing to pay for value created. Parcell is not recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0383	SimWerx, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Appendage-mounted display apparatus</i>
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: SimWerx, Inc. intends to commercialize a forearm-mounted display with features that are crucial for prolonged casualty care for military and civilian first responders.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Three full-time employees of SimWerx (an operating company) cover the required roles and responsibilities. Prior successful product development and fundraising.
R	Opportunity/Market Size	Proposal has no statement or definition of the specific market opportunity. The Market Opportunity section describes very large broad markets, of which this product would be a very small part.
R	Intellectual Property Protection	One issued patent with very narrow claims. No letter of support.
G	Proof of Concept	Working prototype obtained from Air Force previously. Website shows a functioning system using a smart phone display.
R	Potential Investor/ Business Partner Engagement	Investor engagement is vague. Follow-on matching opportunities are clearly stated if lead funding can be raised.
R	Business Model	Sales forecasts appear unrealistic as a result of poor description of market access and penetration.
R	Project Plan/ Budget Narrative	Project plan achievable. Inadequate descriptions of vendors.
Y	Growth Plan in Ohio	Colorado-based with an Ohio facility. Growth in Ohio planned.
Y	ESP Interaction	Recent engagement with the Dayton Entrepreneurs' Center.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Experienced team in an operating, revenue generating company. Company is based in Colorado with a facility in Ohio. The market opportunity for this specific device is not defined. Intellectual property position is thin, and no letter of support was provided. Evidence of access to markets is lacking. Engagement with potential lead investors is not well described. This application is not recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0384	Skuld LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$ 0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	AF96 Evaporative Casting
Advanced Manufacturing		Entrepreneurs' Center

Company Snapshot: Skuld LLC provides a more efficient manufacturing process and more effective patented steel alloy, AF96, that reduces raw material costs by 50% and uses only domestically sourced materials in the production of munitions.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	The team has technical expertise with experience in startup, manufacturing and business. They have successfully obtained support for technology development; future growth will require personnel with fund raising and product manufacturing expertise.
R	Opportunity/Market Size	Skuld is focused on manufacturing Howitzer shell bodies with increased production and economic benefit. Market size is based on DoD goals, further specificity needed to quantify the opportunity.
Y	Intellectual Property Protection	Material for Howitzer bodies is covered by a partially exclusive license agreement held by AFRL. Skuld has existing know-how and patents covering casting technologies to be used.
Y	Proof of Concept	Material production technology has been demonstrated. The proposed work will specifically focus on demonstration with AF96 and provide material to measure key metrics for proof of MVP.
Y	Potential Investor/ Business Partner Engagement	Discussions with ~50 investors and are investigating crowdfunding and non-dilutive funding.
R	Business Model	Initial sales to the US Army will be extended to DoD contractors. Optimistic revenue predictions suggest \$48M in Y5.
Y	Project Plan/ Budget Narrative	Plan is consistent with production of MVP material.
Y	Growth Plan in Ohio	Production facility in Piqua is projected to provide 200 jobs in 5 years.
Y	ESP Interaction	Actively involved with the Dayton Entrepreneurial Center.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The proposed work combines novel material compositions with advanced manufacturing technologies to meet the need for increased production levels of critical munitions for the US Army. The proposers have demonstrated the required technical skills to address many of the issues for successful implementation of the proposed approach. Future growth requires a more detailed definition of the market opportunity and associated profit margins, funds raising to support growth plans, transition to a routine manufacturing business mentality and inclusion in the DOD munitions supply chain. This may require the addition of personnel not currently included in the business proposal.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 25-0385	Unspent	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	<i>Coffee Waste into a High Value Additive for Bioplastics</i>
Advanced Materials		Rev1

Company Snapshot: Unspent has developed a patent-pending process that transforms waste coffee grounds into COPOXEE, a biobased material for producing flexible bioplastics.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Team consists of CEO, CTO, Marketing Manager, and COO with reasonable background and experience relevant to the project scope. Only the COO is full time. Direct experience / working knowledge of the plastic/ chemical industry is lacking.
Y	Opportunity/Market Size	Market for biodegradable and functional bio-plasticizers is in the several hundred million dollars per year and growing. Addressable market size is not given.
Y	Intellectual Property Protection	US18/687,973 has been filed. A patent landscape study has been performed by the legal staff at OSU with a favorable analysis resulting.
Y	Proof of Concept	In a prior Phase I Accelerator Award project, technical feasibility of the bio-plasticizer-- COPOXEE- has been demonstrated. Current TRL is a 6.
Y	Potential Investor/ Business Partner Engagement	Too early for engaging potential investors/ partners. Seems to have relevant contacts to engage after Phase 2 project is successful.
R	Business Model	Reasonable business model. Major question is the estimated selling price for the product at \$ 20/ lb. which is highly unrealistic.
Y	Project Plan/ Budget Narrative	Plan to make 3.5 Kg of product for field testing.
Y	Growth Plan in Ohio	Plan to build or work with OH based cos to produce product and grow
Y	ESP Interaction	Working with local ESP.
	Evaluator Recommendation	This application is not recommended for funding.

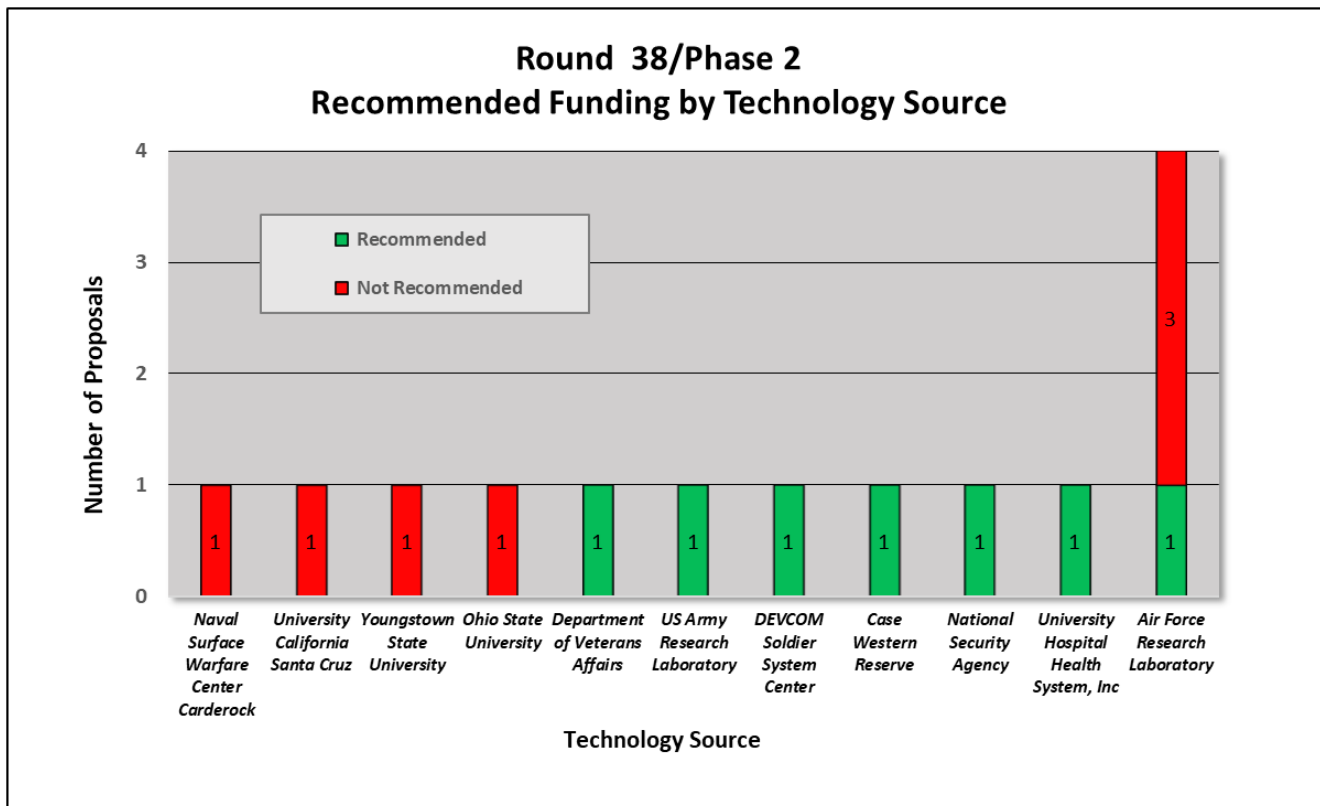
Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The opportunity for using spent coffee ground waste in bio-plasticizers market is very attractive and has merit. However, real opportunity for a high-priced bio-plasticizers in the target markets is very challenging. The Team is encouraged to critically evaluate their process/ manufacturing costs to be more competitive in the market place. It would be helpful to focus on one high priority application and develop the application data in field conditions to show the efficacy of their bio-plasticizers and their acceptance on a cost-performance basis. A revised business model and proforma projection would be helpful. The Team is encouraged to work closely with ESP to further refine their potential partner / customer engagements.

4) Round 38 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. The Air Force Research Laboratory submitted the most proposals with 4, one of which was recommended. Single application submissions were received sourcing technology from Naval Surface Warfare Center Carderock, University California Santa Cruz, Youngstown State University*, Ohio State University, Department of Veterans Affairs, US Army Research Laboratory, DEVCOM Soldier System Center, Case Western Reserve University, National Security Agency and University Hospital Health System, Inc. All of the applications with out of state and AFRL technologies worked with the Entrepreneurs' Center in Dayton. One each from Department of Veterans Affairs, US Army Research Laboratory, DEVCOM Soldier System Center, Case Western Reserve University, National Security Agency, University Hospital Health System, Inc. and Air Force Research Laboratory are recommended for funding.

Figure 1. Round 38 Funding by Technology Source



*Youngstown State University was identified by Impeccable Air as the ESP. The applicant did not meet the TVSF criteria for advancing to the interview stage.

Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In this Round, five of fourteen were in Software/Information Technology (36%), three of the fourteen were in each of Advanced Materials (21%) and Biomedical/Life Sciences (21%), two of the fourteen proposals were in Advanced Manufacturing (14%) and one of the fourteen was in Sensors (7%).

Three of the fourteen in Software/Information Technology (21%), two of the fourteen in Biomedical/Life Sciences (14%), and one of the fourteen in each of Advanced Manufacturing (7%) and Sensors (7%) were recommended for funding. This round represents 21% in Third Frontier Technology areas that were Biomedical/Life Sciences. The previous eighteen rounds have averaged 55% of the applications in Biomedical/Life Sciences.

Figure 2. Round 38 Phase 2 Proposal Activity by Third Frontier Technology Area

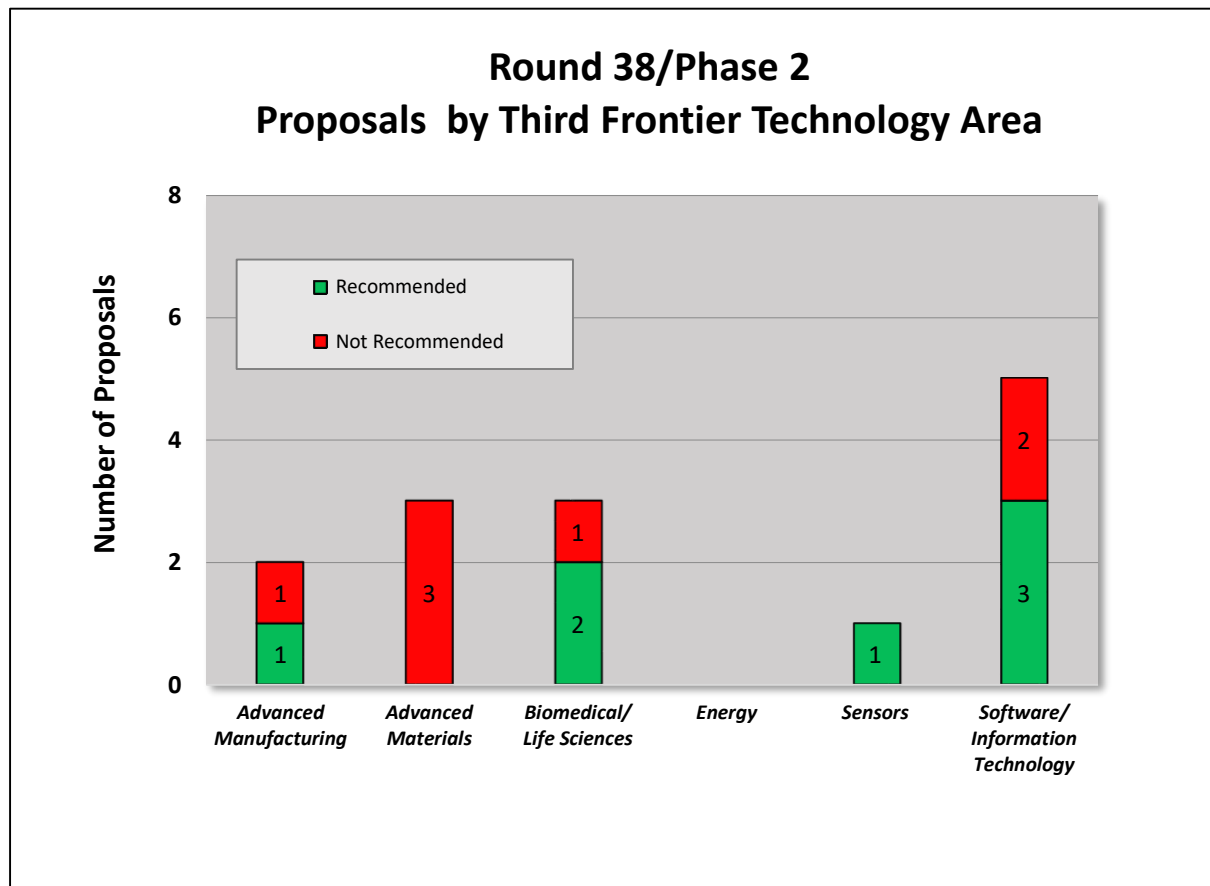
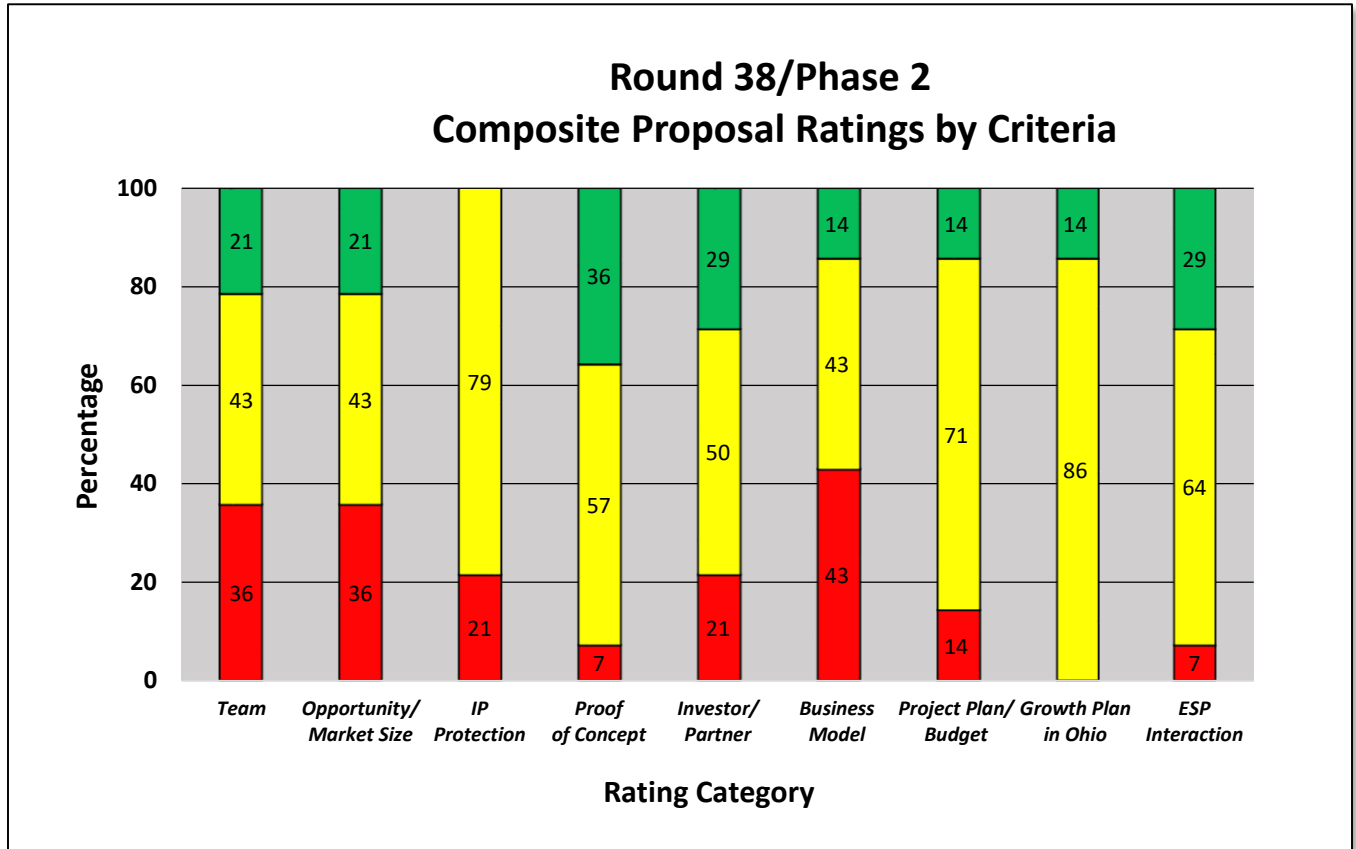


Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. Setting aside the Growth Plan in Ohio category, the Proof of Concept and ESP Interaction had the lowest percentage of weak scores in this round. Business Model followed by Team and Opportunity/Market Size had the highest percentage of weak scores in this round.

Figure 3. Round 38 Phase 2 Proposal Rating Summary



In the previous eighteen Rounds, business model was the lowest rating in Rounds 20-23 (53% average $\geq$ meets) and Round 26 (28%). The RFP was revised prior to Round 24 to elicit stronger business models and it appears that the proposals have provided stronger business models in Rounds 24-37 (66% average $\geq$ meets). Round 26 appears to be an anomaly. Future rounds will be monitored to see if the improvement continues.

Figure 4: Rounds 20 to 38 Phase 2 Analysis of Business Model

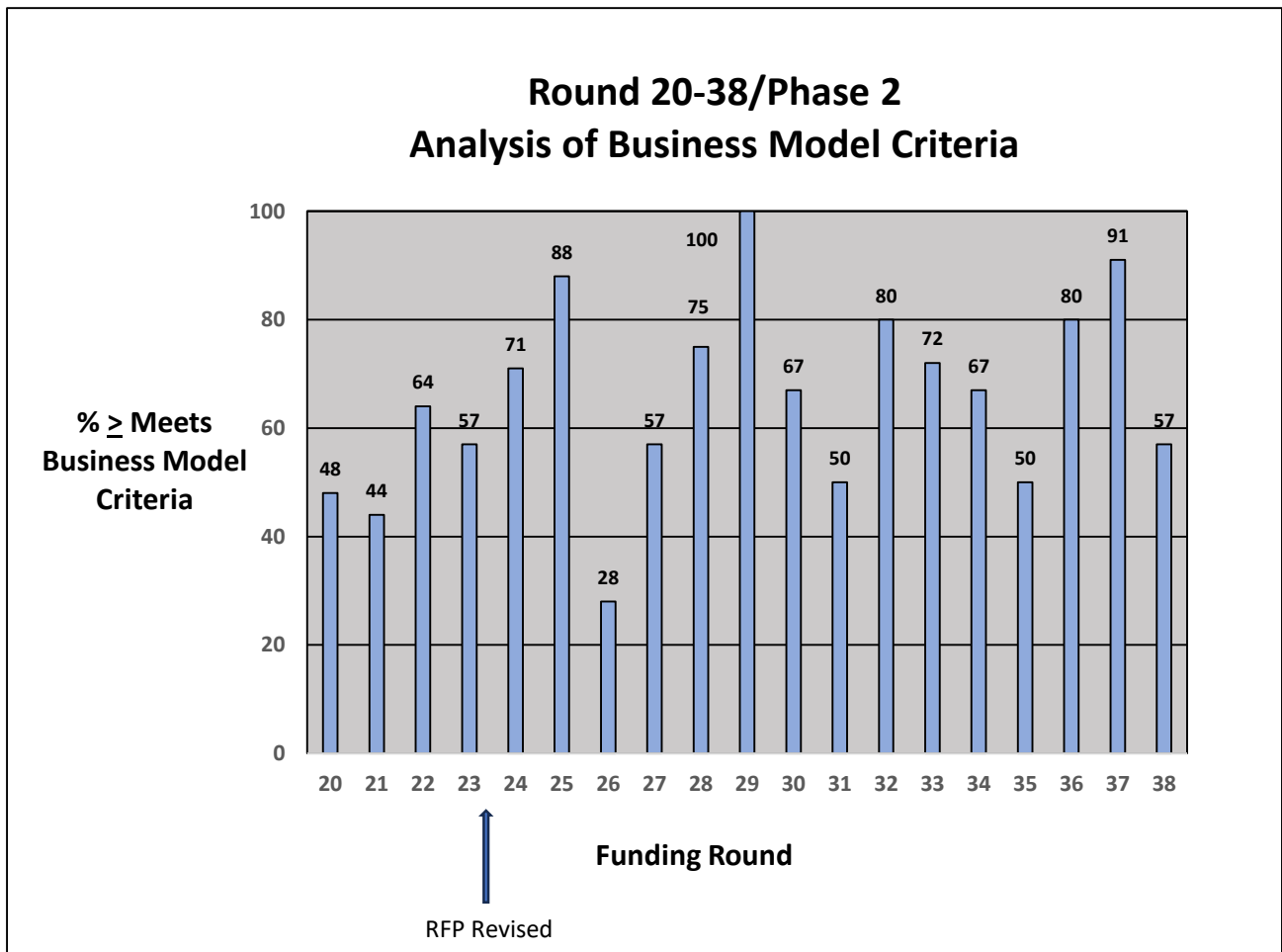
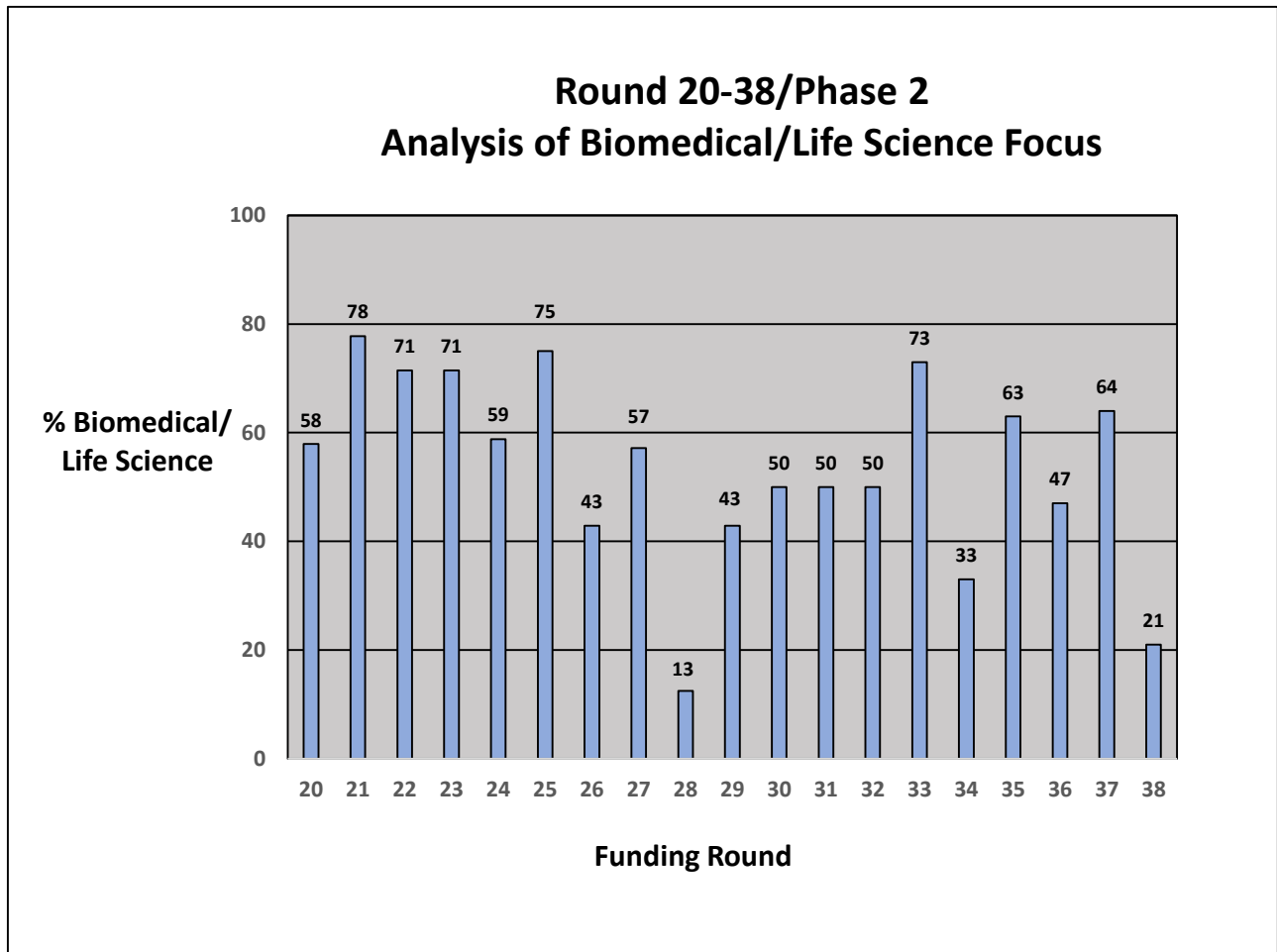


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last nineteen Rounds. Round 38 represents 21% in Third Frontier Technology areas that are Biomedical/Life Sciences, the lowest since Round 28. The primary contributor to this is the large number of applications associated with federal laboratories and out of state universities working with the Entrepreneurs' Center. The previous eighteen rounds average 55% of the applications in Biomedical/Life Sciences.

Figure 5: Rounds 20-38 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There were 2 Phase 2 applicants that previously received Phase 1 funding. One application is recommended for funding.

Phase 2: There were 4 reapplications in this Round. Two were first time reapplications and both are recommended for funding. Two were second time reapplications and one is recommended for funding.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 9 years while each team member has been active in this field for over 25 years.

Each Redwood team member

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The five members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

Market/Technology Assessment (140+ years)

Startup/ Spin out companies (50+ years)

Board member/Advisor to Startups (30+ years)

Evaluating/ monitoring RFPs/ Funding selection (40+ years)

The following is a brief summary of the five principal team members used in this evaluation Round.

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to \$52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

John McArdle

- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance / International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPH, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents

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- Former VP Market and Technology Assessment at the National Technology Transfer Center
- Over 15 years as an angel investor in technology-based startups
- Over 15 years as an evaluator for Ohio Third Frontier funds including IOF, CALF and TIP
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Bhima Vijayendran

- BS, University of Madras, MS, University of Madras, PhD, University of Southern California in Polymer and Surface Science, MBA, University of New Haven
- Former Senior Research Leader and Vice President Business Development, Battelle Memorial Institute; Chief Research Officer, Battelle Science and Technology, Malaysia
- Former Director, Discovery Research, PPG Industries
- Recognized as one of the leading authorities on advanced materials, special chemical and polymer systems in numerous markets including: Renewable and clean technology, Energy, Nano Technology and Industrial Products.
- Recipient of ten R&D 100 awards and over 100 patents and numerous other awards.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/ proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent 3rd party source.

The goals of Phase 2 include:

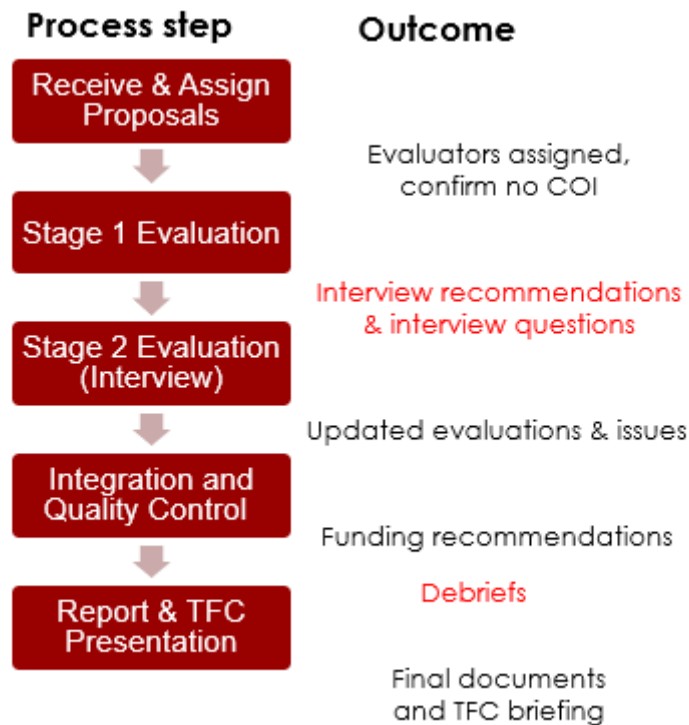
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of life of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation and market research/ business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

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Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person, 45-minute interviews were held with each invited applicant to discuss the advance questions plus other topics of interest to the evaluators. Two Redwood team members participated in the interviews in person with additional team members joining via Zoom or a conference call.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go / No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3 rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

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Table 2 – Phase 2 Evaluation Criteria

Criterion	Weighting	Description
Alignment & compliance	Go / No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity / market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection / License	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition, is license likely within 9 months?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor / Business Partner Engagement	10	Is there company engagement / collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service / manufacturing model be scaled?
Project Plan / Budget Narrative	5	Is the budget consistent with proof in 1 year?
Start-up in Ohio	5	Does a start-up exist or is it planned? Will the start-up be in Ohio?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?